7 (a) (i) Explain how the existence of a threshold frequency in the photoelectric effect provides evidence of the particulate nature of electromagnetic radiation.

.....

.....

.....

..... [2]

(ii) When a beam of electrons is directed at a double slit, an interference pattern is observed on a screen behind the slits. Explain how this observation provides **two** pieces of evidence for the wave nature of electrons.

1 .....

.....

2 .....

..... [2]

(b) An electron is confined along the x-axis between two rigid walls located at $x = 0$ and $x = 3.00$ cm.

(i) Calculate the energy difference between the ground state and the first excited state of the electron.

energy = ... J [2]

(ii) Hence, determine the wavelength of the photon emitted when the electron transitions from the first excited state to the ground state.

wavelength = ... m [2]

(iii) At a particular instant in time, the electron has the following wavefunction:

$$\psi(x) = \begin{cases} A\sin(kx) & \text{for } 0 \le x \le 3.00\,\text{cm} \\ 0 & \text{elsewhere} \end{cases}$$

where A is the normalisation constant and $k = \dfrac{2\pi}{3.00}$ cm^{-1}.

1. In Fig. 7.1 below, sketch the variation of the probability density $|\psi(x)|^2$ with position x. You are not required to label values.

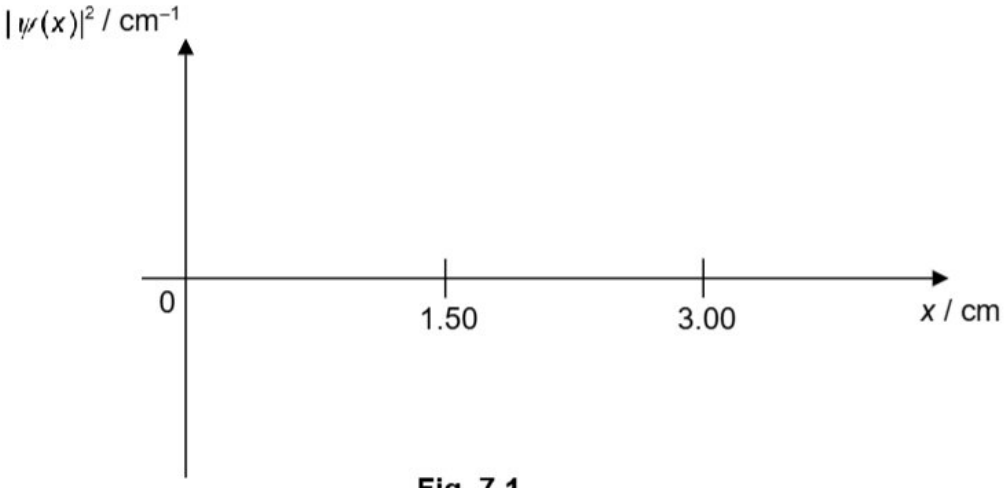


Fig. 7.1

[1]

2. Hence, determine the normalisation constant A.

A = ... cm$^{-\frac{1}{2}}$ [2]

3. Determine the probability of finding the electron between x = 1.50 cm and x = 2.25 cm.

probability = ... [1]